

# SAP on RHEL

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# Home

## Introduction

Course Title: SAP on RHEL

Description: Course description FIXME

Duration: FIXME hours

## Topics covered in this course

- Overview of RHEL for SAP Solutions
- Red Hat Portfolio for SAP
- SAP HANA Infrastructure

## Prerequisites

This course assumes that you have the following prior experience:

- Course or experience...
- Course or experience...
- Course or experience...



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# Lab Environment

FIXME: Instructions for accessing the lab environment for this hands-on based training.

# Overview of RHEL for SAP Solutions

Red Hat Enterprise Linux (RHEL) for SAP Solutions provides a standardized, automated infrastructure designed to support SAP workloads, including SAP S/4HANA, across on-premises and cloud environments.

Key components of the RHEL for SAP Solutions subscription include:

- **RHEL Foundation:** The core enterprise Linux operating system.
- **High Availability:** The High Availability Add-on for maintaining continuous SAP service uptime.
- **Management & Insights:** Tools for lifecycle management (Smart Management), proactive risk analytics, and automated mitigation (Red Hat Insights).
- **Automation:** RHEL System Roles for SAP and integration with Red Hat Ansible Automation Platform to streamline SAP deployments and day-2 operations.
- **SAP Optimization:** Pre-configured SAP-specific libraries and configurations to ensure optimal system performance.
- **Extended Lifecycle:** Update Services for SAP Solutions, providing support for specific RHEL minor releases for up to four years.

## Introduction to Red Hat Enterprise Linux for SAP

Red Hat Enterprise Linux (RHEL) for SAP Solutions is a purpose-built operating system environment designed to meet the rigorous performance, stability, and reliability requirements of SAP workloads. As organizations migrate to SAP S/4HANA or modernize their existing Digital Core, RHEL serves as the foundation for an Open Hybrid Cloud strategy, allowing for consistent operations across on-premises data centers and public cloud providers.

### Why RHEL for SAP?

SAP applications are mission-critical, often serving as the central nervous system of an enterprise. Deploying these workloads on standard Linux distributions can lead to performance bottlenecks or lack of specialized support. RHEL for SAP Solutions addresses these challenges by bundling critical components specifically tuned for the SAP ecosystem.

Unlike a generic OS, the RHEL for SAP Solutions subscription provides an automated, end-to-end infrastructure specifically engineered to minimize downtime, simplify management, and ensure compliance with SAP's technical requirements.

### Key Components of the RHEL for SAP Portfolio

The RHEL for SAP Solutions subscription is more than just an OS; it is a comprehensive suite of tools designed to manage the entire lifecycle of an SAP environment:

- **Red Hat Enterprise Linux:** The core, stable foundation optimized for high-performance enterprise applications.

- **High Availability (HA) Add-on:** Provides the clustering framework necessary to ensure SAP services (like SAP HANA or SAP NetWeaver) remain operational during hardware or service failures.
- **Red Hat Smart Management:** Simplifies lifecycle management, allowing administrators to manage patch levels and configurations across large-scale SAP environments.
- **Red Hat Insights:** An analytics-driven tool that provides proactive optimization, identifies security risks, and suggests automated mitigations before they impact production performance.
- **RHEL System Roles for SAP:** Pre-defined automation roles that enforce consistent configurations (such as kernel parameters and storage tuning) to match SAP best practices during deployment.
- **SAP-specific Software Components:** Optimized libraries and configurations tailored for SAP workload requirements.
- **Update Services for SAP Solutions:** Offers extended support for specific minor RHEL releases—four years from General Availability—enabling SAP customers to maintain stable operating environments while managing longer update cycles.

## Automation with Ansible

Beyond the OS components, Red Hat facilitates the full automation of SAP S/4HANA migrations and Day-2 operations through the integration with the **Red Hat Ansible Automation Platform**. By combining RHEL for SAP Solutions with Ansible, IT teams can transform manual, error-prone deployment processes into repeatable, standardized workflows, significantly reducing the "time-to-value" for new SAP initiatives.



For organizations scaling their SAP footprint, leveraging **Red Hat Insights** is crucial. It acts as a continuous health-check tool, mapping your RHEL configuration directly against known SAP support notes and security advisories to prevent potential outages.

## Hands-on Activity: Identifying SAP-Specific Components

In this activity, you will verify if the necessary SAP-specific repositories are enabled on your RHEL system.

### 1. Prerequisites

- A system registered with a valid RHEL for SAP Solutions subscription.
- Root or **sudo** access.

### 2. Step 1: Check enabled repositories

Run the following command to view your currently enabled repositories and ensure the SAP-specific repos are active:

```
subscription-manager repos --list-enabled | grep -i sap
```

### 1. Step 2: Verify SAP-tuned profiles

RHEL for SAP includes pre-configured **tuned** profiles. Check which profiles are available for your environment:

```
tuned-adm list
```

Look for profiles prefixed with **sap-** (e.g., **sap-hana** or **sap-netweaver**). These profiles automatically adjust kernel parameters, disk I/O schedulers, and memory settings to align with SAP's performance requirements.

1. **Step 3: Apply an SAP profile** If your server is hosting an SAP HANA database, apply the SAP HANA tuned profile:

```
sudo tuned-adm profile sap-hana
```

By executing these steps, you have ensured that your OS is not just running, but is specifically optimized for the SAP HANA workload requirements.

## Understanding the RHEL for SAP Solutions subscription

The RHEL for SAP Solutions subscription is more than just an operating system license; it is an integrated, specialized platform designed to meet the rigorous performance, reliability, and lifecycle requirements of SAP environments. By consolidating essential components into a single subscription, Red Hat enables organizations to build a standardized, automated, and resilient foundation for their SAP Digital Core.

### Subscription Components

The RHEL for SAP Solutions subscription provides a comprehensive bundle of software and services designed to streamline the deployment and management of SAP workloads. Key components include:

- **Red Hat Enterprise Linux (RHEL):** The core, hardened Linux distribution that serves as the foundation for SAP NetWeaver and SAP HANA workloads.
- **High Availability (HA) Add-on:** A robust clustering framework specifically engineered to support SAP-certified high-availability configurations, ensuring minimal downtime for critical SAP services.
- **Red Hat Smart Management:** Provides integrated lifecycle management, allowing administrators to manage software patches and security updates across large-scale RHEL deployments.
- **Red Hat Insights:** A predictive analytics tool that continuously monitors system health, offering proactive risk assessment and automated mitigation strategies to identify potential issues before they impact business operations.
- **RHEL System Roles for SAP:** A collection of Ansible-based roles that automate the complex configuration tasks required to prepare a RHEL environment for SAP applications, adhering to SAP-specific best practices.

- **SAP-Specific Software Components:** Includes pre-configured libraries and kernel tuning parameters required to meet the SAP "Tuning Guide" requirements for optimal performance.
- **Update Services for SAP Solutions (E4S):** A critical feature for SAP environments that provides long-term support for specific minor RHEL releases (for four years from General Availability), allowing for more flexible maintenance windows and reduced update frequency.

## Value Proposition: The Open Hybrid Cloud

By leveraging this subscription, organizations can execute an Open Hybrid Cloud strategy. Because the subscription is consistent across on-premises, private, and public cloud environments, SAP customers can:

1. **Standardize infrastructure:** Use the same RHEL foundation regardless of the deployment site.
2. **Automate operations:** Integrate with Red Hat Ansible Automation Platform to orchestrate SAP S/4HANA migrations and day-2 operations.
3. **Ensure Compliance:** Utilize the included management tools to maintain consistency across the entire SAP landscape.

### Hands-on Activity: Verifying Subscription Benefits

In this exercise, you will confirm your system is correctly registered and configured to access the SAP-specific repositories provided by your subscription.



Ensure your system is registered with Red Hat Subscription Management (RHSM) before beginning this activity.

1. **Check Registration Status** Execute the following command to verify your system is registered and the subscription is attached: [source,bash] `--- subscription-manager status ---`
2. **Enable SAP-Specific Repositories** The subscription provides specialized repositories. Verify and enable the required repositories for your RHEL version: [source,bash] `--- # List available repositories to find the SAP-specific ones subscription-manager repos --list | grep -i "sap"`

```
# Enable the RHEL for SAP Solutions repository (example for RHEL 8)
subscription-manager repos --enable=rhel-8-for-x86_64-sap-solutions-rpms
----
```

3. **Verify Insights Connectivity** To leverage the proactive optimization features of your subscription, ensure the Red Hat Insights client is active: [source,bash] `--- insights-client --register ---`
4. **Validate Automation Readiness** Confirm that the `rhel-system-roles` package is installed to begin automating your SAP configurations: [source,bash] `--- dnf install rhel-system-roles ---`

By completing these steps, you have ensured that your environment is properly configured to leverage the support and automation capabilities unique to the RHEL for SAP Solutions subscription.

# Red Hat Portfolio for SAP

The Red Hat portfolio provides an automated, end-to-end open hybrid cloud infrastructure designed to support SAP and non-SAP workloads across on-premises and cloud environments. It enables organizations to standardize their SAP S/4HANA deployments on a reliable platform.

Key components of the portfolio include:

- **RHEL for SAP Solutions:** A bundled subscription including the base OS, High Availability Add-on, Smart Management, and Red Hat Insights for proactive risk mitigation.
- **Automation and Lifecycle:** Utilizes RHEL System Roles for SAP and Red Hat Ansible Automation Platform to automate migrations, deployments, and day-2 operations.
- **SAP-Specific Optimization:** Includes specialized configurations, libraries, and update services that provide extended support for specific minor RHEL releases to ensure stability for SAP environments.

## Open Hybrid Cloud strategy for SAP workloads

### Overview

In the modern enterprise landscape, SAP environments are no longer confined to static, on-premises data centers. To support the "Intelligent Enterprise," organizations must adopt a flexible, scalable, and resilient infrastructure. Red Hat's Open Hybrid Cloud strategy provides a consistent foundation for SAP workloads, allowing businesses to move seamlessly between on-premises environments and public cloud providers while maintaining operational consistency.

### The Red Hat Approach to SAP Workloads

Red Hat enables SAP customers to deploy a unified, automated, and end-to-end infrastructure. Whether you are performing a greenfield deployment of SAP S/4HANA or migrating legacy systems, the Open Hybrid Cloud strategy ensures that your "Digital Core"—comprising SAP S/4HANA, BW/4HANA, and SAP HANA databases—operates with the same reliability and management standards across every footprint.

Key components of this strategy include:

- **Operational Consistency:** Use the same Red Hat Enterprise Linux (RHEL) foundation across on-premises servers, private clouds, and major public cloud providers (AWS, Azure, Google Cloud).
- **Automation:** Utilize Red Hat Ansible Automation Platform to standardize SAP installation, configuration, and maintenance, reducing human error and deployment time.
- **Scalability:** Rapidly provision infrastructure to match business growth or seasonal peaks in SAP system demand.
- **High Availability:** Implement enterprise-grade clustering solutions to ensure that critical SAP business processes remain online, even in the event of hardware or software failure.

## Hands-on Activity: Planning a Multi-Cloud SAP Deployment

This activity guides you through the high-level planning required to support a hybrid architecture for SAP.

### Prerequisites

- Access to the Red Hat Customer Portal.
- Access to the SAP Service Marketplace (for SAP software binaries).
- A base understanding of your organization's compliance requirements for data locality.

### Step 1: Inventory Assessment

Before deploying, categorize your SAP components. Identify which workloads are: 1. **Core Systems (S/4HANA)**: These typically require high-performance, low-latency, and high-availability (HA) setups. 2. **Modern Cloud-Native Apps**: IoT, Business Networks, and Customer Experience applications that may run on containers.

### Step 2: Selecting the Infrastructure footprint

Define where your systems will reside. For a true Hybrid Cloud strategy: \* **On-Premises**: Deploy RHEL for SAP Solutions to leverage existing hardware investments. \* **Public Cloud**: Utilize the pre-configured "SAP-certified" images available in cloud marketplaces to ensure full support and performance optimization.

### Step 3: Standardizing with Automation

Create a standard baseline using Ansible Playbooks to ensure that every RHEL instance—regardless of location—has the necessary tuning for SAP HANA: \* Ensure **tuned** profiles (e.g., **sap-hana**) are applied. \* Verify that required kernel parameters are set to SAP-compliant values.

```
# Example: Apply the SAP HANA tuned profile to your RHEL instance
tuned-adm profile sap-hana
```

### Step 4: Verification

Confirm that your hybrid footprint is reporting correctly to your management consoles (such as Red Hat Insights for RHEL). This ensures that you have visibility into the health and security of your SAP estate, regardless of whether it is running on-premises or in the cloud.

## Troubleshooting Guidance

- **Consistency Issues**: If an SAP system behaves differently in the cloud compared to on-premises, verify that the same RHEL for SAP Solutions repositories are being used. Cloud-specific repositories often contain specialized drivers and optimization packages required for SAP performance.
- **High Availability**: Ensure that your cluster fencing (STONITH) mechanisms are configured

according to the specific requirements of your cloud provider, as these differ significantly from traditional hardware-based IPMI/ILO fencing.

# SAP HANA Infrastructure

SAP HANA infrastructure on RHEL serves as the foundation for modern SAP S/4HANA environments. Key architectural considerations include:

- **Deployment Models:** Support for both scale-up and scale-out configurations based on performance and data volume requirements.
- **High Availability:** Integration with RHEL HA solutions, such as Pacemaker, to automate failover processes and ensure service continuity.
- **System Replication (HSR):** Implementation of HANA System Replication across nodes to maintain data synchronization and enable rapid disaster recovery.
- **RHEL Optimization:** Leveraging specialized RHEL for SAP Solutions subscriptions to meet the performance, security, and reliability requirements of the SAP kernel and database layer.
- **Standardized Setup:** Adherence to identical node configurations to facilitate seamless clustering and environment management.

## SAP HANA on RHEL Architecture Requirements

Running SAP HANA on Red Hat Enterprise Linux (RHEL) requires strict adherence to certified configurations to ensure performance, stability, and supportability. This section outlines the architectural requirements and foundational considerations for deploying SAP HANA workloads.

### Prerequisites and Supportability

Before initiating a deployment, it is critical to align the infrastructure with both Red Hat and SAP guidelines.



In official documentation, the term **RHEL** serves as a synonym for both **RHEL for SAP HANA** and **RHEL for SAP Solutions**. Always consult the [Certified and Supported SAP HANA Hardware Directory](#) to ensure your hardware choice is validated for your specific workload.

### Hardware Certification

SAP HANA requires specific hardware profiles to handle high-memory throughput and low-latency storage requirements. \* **Certified Hardware:** You must use servers, storage systems, and cloud instance types that are explicitly listed in the SAP HANA Hardware Directory. \* **Verification:** For non-appliance scenarios, verify that your specific server model is compatible with your RHEL release using the [Red Hat Hardware Compatibility List](#).

### Deployment Models

Architectural requirements differ based on the deployment scale: \* **Scale-Up:** A single-node environment where the database is contained within one server. \* **Scale-Out:** A distributed environment where the database is partitioned across multiple nodes.

For both models, if implementing **High Availability (HA)**, you must utilize the **RHEL High Availability Add-On**. Support for these environments follows the specific policies defined in the [Support Policies for RHEL High Availability Clusters](#).

## Hands-on Activity: Verifying System Readiness

This activity guides you through the process of verifying that your RHEL environment meets the basic criteria for an SAP HANA installation.

### Prerequisites

- An active RHEL for SAP Solutions subscription.
- Root access to the target RHEL server.

### Step 1: Check RHEL Subscription Type

Ensure your system is registered with the correct subscription, which provides access to the necessary repositories for SAP HANA.

```
# Verify the active subscription
subscription-manager list --consumed
```

### Step 2: Validate Hardware Certification

Use the following command to retrieve your hardware vendor and product information to compare against the SAP HANA Hardware Directory:

```
# Retrieve hardware information
dmidecode -t system | grep -E "Manufacturer|Product"
```

### Step 3: Check Repository Availability

Confirm that the SAP-specific repositories are enabled to access the necessary packages for SAP HANA optimizations.

```
# List enabled repositories for SAP
yum repolist | grep sap
```



If the repositories are missing, use `subscription-manager repos --enable=rhel-sap-for-rhel-8-x86_64-rpms` (adjust for your RHEL version) to enable the required channels.

## Architectural Best Practices

- **Update Services:** Leverage the **Update Services for SAP Solutions (E4S)**. This allows you to

stay on a specific RHEL minor release for an extended period, which is crucial for the lifecycle management of SAP HANA environments. Refer to the [Red Hat support policy for Update Services](#) for details.

- **HA Implementation:** When designing for HA, ensure that your fencing mechanisms and resource agents are configured according to the Red Hat official documentation for SAP HANA clusters to prevent data corruption during failover events.

# Appendix

FIXME: Content for Appendix...

## Resources

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